AES Example - Input (128 bit key and message)

Key in English: Thats my Kung Fu (16 ASCII characters, 1 byte each)

Translation into Hex:

T	h	a	t	s		m	y		K	u	n	g		F	u
54	68	61	74	73	20	6D	79	20	4B	75	6E	67	20	46	75

Key in Hex (128 bits): 54 68 61 74 73 20 6D 79 20 4B 75 6E 67 20 46 75

Plaintext in English: Two One Nine Two (16 ASCII characters, 1 byte each)

Translation into Hex:

T	w	o		O	n	e		N	i	n	e		T	w	o
54	77	6F	20	4F	6E	65	20	4E	69	6E	65	20	54	77	6F

Plaintext in Hex (128 bits): 54 77 6F 20 4F 6E 65 20 4E 69 6E 65 20 54 77 6F

AES Example - The first Roundkey

- Key in Hex (128 bits): 54 68 61 74 73 20 6D 79 20 4B 75 6E 67 20 46 75
- $w[0] = (54, 68, 61, 74), w[1] = (73, 20, 6D, 79), w[2] = (20, 4B, 75, 6E), w[3] = (67, 20, 46, 75)$
- $g(w[3])$:
  - circular byte left shift of $w[3]$: $(20, 46, 75, 67)$
  - Byte Substitution (S-Box): $(B7, 5A, 9D, 85)$
  - Adding round constant $(01, 00, 00, 00)$ gives: $g(w[3]) = (B6, 5A, 9D, 85)$
- $w[4] = w[0] \oplus g(w[3]) = (E2, 32, FC, F1)$:

0101 0100 1011 0110	0110 1000 0101 1010	0110 0001 1001 1101	0111 0100 1000 0101
1110 0010 E2	0011 0010 32	1111 1100 FC	1111 0001 F1

- $w[5] = w[4] \oplus w[1] = (91, 12, 91, 88)$, $w[6] = w[5] \oplus w[2] = (B1, 59, E4, E6)$,
  $w[7] = w[6] \oplus w[3] = (D6, 79, A2, 93)$
- first roundkey: E2 32 FC F1 91 12 91 88 B1 59 E4 E6 D6 79 A2 93

AES Example - All RoundKeys

- Round 0: 54 68 61 74 73 20 6D 79 20 4B 75 6E 67 20 46 75
- Round 1: E2 32 FC F1 91 12 91 88 B1 59 E4 E6 D6 79 A2 93
- Round 2: 56 08 20 07 C7 1A B1 8F 76 43 55 69 A0 3A F7 FA
- Round 3: D2 60 0D E7 15 7A BC 68 63 39 E9 01 C3 03 1E FB
- Round 4: A1 12 02 C9 B4 68 BE A1 D7 51 57 A0 14 52 49 5B
- Round 5: B1 29 3B 33 05 41 85 92 D2 10 D2 32 C6 42 9B 69
- Round 6: BD 3D C2 B7 B8 7C 47 15 6A 6C 95 27 AC 2E 0E 4E

8

the PDF has a typo there.

- Round 7: CC 96 ED 16 74 EA AA 03 1E 86 3F 24 B2 A8 31 6A
- Round 8: 8E 51 EF 21 FA BB 45 22 E4 3D 7A 06 56 95 4B 6C
- Round 9: BF E2 BF 90 45 59 FA B2 A1 64 80 B4 F7 F1 CB D8
- Round 10: 28 FD DE F8 6D A4 24 4A CC C0 A4 FE 3B 31 6F 26



AES Example - Add Roundkey, Round 0

- State Matrix and Roundkey No.0 Matrix:

  from plaintxt

$$\begin{pmatrix} 54 & 4F & 4E & 20 \\ 77 & 6E & 69 & 54 \\ 6F & 65 & 6E & 77 \\ 20 & 20 & 65 & 6F \end{pmatrix} \qquad \begin{pmatrix} 54 & 73 & 20 & 67 \\ 68 & 20 & 4B & 20 \\ 61 & 6D & 75 & 46 \\ 74 & 79 & 6E & 75 \end{pmatrix}$$

- XOR the corresponding entries, e.g., $69 \oplus 4B = 22$

$$\begin{array}{r} 0110\ 1001 \\ 0100\ 1011 \\ \hline 0010\ 0010 \end{array}$$

- the new State Matrix is

$$\begin{pmatrix} 00 & 3C & 6E & 47 \\ 1F & 4E & 22 & 74 \\ 0E & 08 & 1B & 31 \\ 54 & 59 & 0B & 1A \end{pmatrix}$$

AES Example - Round 1, Substitution Bytes

- current State Matrix is

$$\begin{pmatrix} 00 & 3C & 6E & 47 \\ 1F & 4E & 22 & 74 \\ 0E & 08 & 1B & 31 \\ 54 & 59 & 0B & 1A \end{pmatrix}$$

- substitute each entry (byte) of current state matrix by corresponding entry in AES S-Box
- for instance: byte 6E is substituted by entry of S-Box in row 6 and column E, i.e., by 9F
- this leads to new State Matrix

$$\begin{pmatrix} 63 & EB & 9F & A0 \\ C0 & 2F & 93 & 92 \\ AB & 30 & AF & C7 \\ 20 & CB & 2B & A2 \end{pmatrix}$$

- this non-linear layer is for resistance to differential and linear cryptanalysis attacks

AES Example - Round 1, Shift Row

- the current State Matrix is

$$\begin{pmatrix} 63 & EB & 9F & A0 \\ C0 & 2F & 93 & 92 \\ AB & 30 & AF & C7 \\ 20 & CB & 2B & A2 \end{pmatrix}$$

- four rows are shifted cyclically to the left by offsets of 0,1,2, and 3
- the new State Matrix is

$$\begin{pmatrix} 63 & EB & 9F & A0 \\ 2F & 93 & 92 & C0 \\ AF & C7 & AB & 30 \\ A2 & 20 & CB & 2B \end{pmatrix}$$

- this linear mixing step causes diffusion of the bits over multiple rounds

AES Example - Round 1, Mix Column

- Mix Column multiplies fixed matrix against current State Matrix:

$$\begin{pmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{pmatrix} \begin{pmatrix} 63 & EB & 9F & A0 \\ 2F & 93 & 92 & C0 \\ AF & C7 & AB & 30 \\ A2 & 20 & CB & 2B \end{pmatrix} = \begin{pmatrix} BA & 84 & E8 & 1B \\ 75 & A4 & 8D & 40 \\ F4 & 8D & 06 & 7D \\ 7A & 32 & 0E & 5D \end{pmatrix}$$

- entry BA is result of $(02 \bullet 63) \oplus (03 \bullet 2F) \oplus (01 \bullet AF) \oplus (01 \bullet A2)$:
  - $02 \bullet 63 = 00000010 \bullet 01100011 = 11000110$
  - $03 \bullet 2F = (02 \bullet 2F) \oplus 2F = (00000010 \bullet 00101111) \oplus 00101111 = 01110001$
  - $01 \bullet AF = AF = 10101111$ and $01 \bullet A2 = A2 = 10100010$
  - hence

$$\begin{array}{r} 11000110 \\ 01110001 \\ 10101111 \\ 10100010 \\ \hline 10111010 \end{array}$$

AES Example - Add Roundkey, Round 1

- State Matrix and Roundkey No.1 Matrix:

$$\begin{pmatrix} BA & 84 & E8 & 1B \\ 75 & A4 & 8D & 40 \\ F4 & 8D & 06 & 7D \\ 7A & 32 & 0E & 5D \end{pmatrix} \qquad \begin{pmatrix} E2 & 91 & B1 & D6 \\ 32 & 12 & 59 & 79 \\ FC & 91 & E4 & A2 \\ F1 & 88 & E6 & 93 \end{pmatrix}$$

- XOR yields new State Matrix

$$\begin{pmatrix} 58 & 15 & 59 & CD \\ 47 & B6 & D4 & 39 \\ 08 & 1C & E2 & DF \\ 8B & BA & E8 & CE \end{pmatrix}$$

- AES output after Round 1: 58 47 08 8B 15 B6 1C BA 59 D4 E2 E8 CD 39 DF CE

AES Example - Round 2

- after Substitute Byte and after Shift Rows:

$$\begin{pmatrix} 6A & 59 & CB & BD \\ A0 & 4E & 48 & 12 \\ 30 & 9C & 98 & 9E \\ 3D & F4 & 9B & 8B \end{pmatrix} \qquad \begin{pmatrix} 6A & 59 & CB & BD \\ 4E & 48 & 12 & A0 \\ 98 & 9E & 30 & 9B \\ 8B & 3D & F4 & 9B \end{pmatrix}$$

- after Mixcolumns and after Roundkey:

$$\begin{pmatrix} 15 & C9 & 7F & 9D \\ CE & 4D & 4B & C2 \\ 89 & 71 & BE & 88 \\ 65 & 47 & 97 & CD \end{pmatrix} \qquad \begin{pmatrix} 43 & 0E & 09 & 3D \\ C6 & 57 & 08 & F8 \\ A9 & C0 & EB & 7F \\ 62 & C8 & FE & 37 \end{pmatrix}$$

AES Example - Round 3

- after Substitute Byte and after Shift Rows:

$$\begin{pmatrix} 1A & AB & 01 & 27 \\ B4 & 5B & 30 & 41 \\ D3 & BA & E9 & D2 \\ AA & E8 & BB & 9A \end{pmatrix} \qquad \begin{pmatrix} 1A & AB & 01 & 27 \\ 5B & 30 & 41 & B4 \\ E9 & D2 & D3 & BA \\ A9 & AA & E8 & BB \end{pmatrix}$$

- after Mixcolumns and after Roundkey:

$$\begin{pmatrix} AA & 65 & FA & 88 \\ 16 & 0C & 05 & 3A \\ 3D & C1 & DE & 2A \\ B3 & 4B & 5A & 0A \end{pmatrix} \qquad \begin{pmatrix} 78 & 70 & 99 & 4B \\ 76 & 76 & 3C & 39 \\ 30 & 7D & 37 & 34 \\ 54 & 23 & 5B & F1 \end{pmatrix}$$

AES Example - Round 4

- after Substitute Byte and after Shift Rows:

$$\begin{pmatrix} BC & 51 & EE & B3 \\ 38 & 38 & EB & 12 \\ 04 & FF & 9A & 18 \\ 20 & 26 & 39 & A1 \end{pmatrix} \qquad \begin{pmatrix} BC & 51 & EE & B3 \\ 38 & EB & 12 & 38 \\ 9A & 18 & 04 & FF \\ A1 & 20 & 26 & 39 \end{pmatrix}$$

- after Mixcolumns and after Roundkey:

$$\begin{pmatrix} 10 & BC & D3 & F3 \\ D8 & 94 & E0 & E0 \\ 53 & EA & 9E & 25 \\ 24 & 40 & 73 & 7B \end{pmatrix} \qquad \begin{pmatrix} B1 & 08 & 04 & E7 \\ CA & FC & B1 & B2 \\ 51 & 54 & C9 & 6C \\ ED & E1 & D3 & 20 \end{pmatrix}$$

AES Example - Round 5

- after Substitute Byte and after Shift Rows:

$$\begin{pmatrix} C8 & 30 & F2 & 94 \\ 74 & B0 & C8 & 37 \\ D1 & 20 & DD & 50 \\ 55 & F8 & 66 & B7 \end{pmatrix} \qquad \begin{pmatrix} C8 & 30 & F2 & 94 \\ B0 & C8 & 37 & 74 \\ DD & 50 & D1 & 20 \\ B7 & 55 & F8 & 66 \end{pmatrix}$$

- after Mixcolumns and after Roundkey:

$$\begin{pmatrix} 2A & 26 & 8F & E9 \\ 78 & 1E & 0C & 7A \\ 1B & A7 & 6F & 0A \\ 5B & 62 & 00 & 3F \end{pmatrix} \qquad \begin{pmatrix} 9B & 23 & 5D & 2F \\ 51 & 5F & 1C & 38 \\ 20 & 22 & BD & 91 \\ 68 & F0 & 32 & 56 \end{pmatrix}$$

AES Example - Round 6

- after Substitute Byte and after Shift Rows:

$$\begin{pmatrix} 14 & 26 & 4C & 15 \\ D1 & CF & 9C & 07 \\ B7 & 93 & 7A & 81 \\ 45 & 8C & 23 & B1 \end{pmatrix} \qquad \begin{pmatrix} 14 & 26 & 4C & 15 \\ CF & 9C & 07 & D1 \\ 7A & 81 & B7 & 93 \\ B1 & 45 & 8C & 23 \end{pmatrix}$$

- after Mixcolumns and after Roundkey:

$$\begin{pmatrix} A9 & 37 & AA & F2 \\ AE & D8 & 0C & 21 \\ E7 & 6C & B1 & 9C \\ F0 & FD & 67 & 3B \end{pmatrix} \qquad \begin{pmatrix} 14 & 8F & C0 & 5E \\ 93 & A4 & 60 & 0F \\ 25 & 2B & 24 & 92 \\ 77 & E8 & 40 & 75 \end{pmatrix}$$

AES Example - Round 7

- after Substitute Byte and after Shift Rows:

$$\begin{pmatrix} FA & 73 & BA & 58 \\ DC & 49 & D0 & 76 \\ 3F & F1 & 36 & 4F \\ F5 & 9B & 09 & 9D \end{pmatrix} \qquad \begin{pmatrix} FA & 73 & BA & 58 \\ 49 & D0 & 76 & DC \\ 36 & 4F & 3F & F1 \\ 9D & F5 & 9B & 09 \end{pmatrix}$$

- after Mixcolumns and after Roundkey:

$$\begin{pmatrix} 9F & 37 & 51 & 37 \\ AF & EC & 8C & FA \\ 63 & 39 & 04 & 66 \\ 4B & FB & B1 & D7 \end{pmatrix} \qquad \begin{pmatrix} 53 & 43 & 4F & 85 \\ 39 & 06 & 0A & 52 \\ 8E & 93 & 3B & 57 \\ 5D & F8 & 95 & BD \end{pmatrix}$$

AES Example - Round 8

- after Substitute Byte and after Shift Rows:

$$\begin{pmatrix} ED & 1A & 84 & 97 \\ 12 & 6F & 67 & 00 \\ 19 & DC & E2 & 5B \\ 4C & 41 & 2A & 7A \end{pmatrix} \qquad \begin{pmatrix} ED & 1A & 84 & 97 \\ 6F & 67 & 00 & 12 \\ E2 & 5B & 19 & DC \\ 7A & 4C & 41 & 2A \end{pmatrix}$$

- after Mixcolumns and after Roundkey:

$$\begin{pmatrix} E8 & 8A & 4B & F5 \\ 74 & 75 & EE & E6 \\ D3 & 1F & 75 & 58 \\ 55 & 8A & 0C & 38 \end{pmatrix} \qquad \begin{pmatrix} 66 & 70 & AF & A3 \\ 25 & CE & D3 & 73 \\ 3C & 5A & 0F & 13 \\ 74 & A8 & 0A & 54 \end{pmatrix}$$

AES Example - Round 9

- after Substitute Byte and after Shift Rows:

$$\begin{pmatrix} 33 & 51 & 79 & 0A \\ 3F & 8B & 66 & 8F \\ EB & BE & 76 & 7D \\ 92 & C2 & 67 & 20 \end{pmatrix} \qquad \begin{pmatrix} 33 & 51 & 79 & 0A \\ 8B & 66 & 8F & 3F \\ 76 & 7D & EB & BE \\ 20 & 92 & C2 & 67 \end{pmatrix}$$

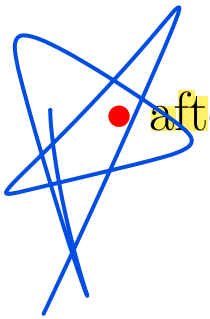
- after Mixcolumns and after Roundkey:

$$\begin{pmatrix} B6 & E7 & 51 & 8C \\ 84 & 88 & 98 & CA \\ 34 & 60 & 66 & FB \\ E8 & D7 & 70 & 51 \end{pmatrix} \qquad \begin{pmatrix} 09 & A2 & F0 & 7B \\ 66 & D1 & FC & 3B \\ 8B & 9A & E6 & 30 \\ 78 & 65 & C4 & 89 \end{pmatrix}$$

AES Example - Round 10

- after Substitute Byte and after Shift Rows:

$$\begin{pmatrix} 01 & 3A & 8C & 21 \\ 33 & 3E & B0 & E2 \\ 3D & B8 & 8E & 04 \\ BC & 4D & 1C & A7 \end{pmatrix} \qquad \begin{pmatrix} 01 & 3A & 8C & 21 \\ 3E & B0 & E2 & 33 \\ 8E & 04 & 3D & B8 \\ A7 & BC & 4D & 1C \end{pmatrix}$$



- after Roundkey (Attention: no Mix columns in last round):

$$\begin{pmatrix} 29 & 57 & 40 & 1A \\ C3 & 14 & 22 & 02 \\ 50 & 20 & 99 & D7 \\ 5F & F6 & B3 & 3A \end{pmatrix}$$

- ciphertext: 29 C3 50 5F 57 14 20 F6 40 22 99 B3 1A 02 D7 3A

ACTUALLy it should be
29 c3 50 5f 57 14 20 f6 40 22 99 b3
1a 02 d7 3a b3 e4 6f 11 ba 8d 2b
97 c1 87 69 44 9a 89 e8 68

demonstrating one AES block only. (didnt do the concatenation here at the END noted)